1. 목적

이 규격은 자동차 부품의 재질표기에 대해서 규정한다. 이 규격의 적용 목적은 자동차의 폐차단계에서 부품의 식별성을 높여 재활용 재질과 폐기 재질의 구분을 용이하게 하며 동일한 재질에 대한 이종 표기를 방지하기 위함이다.

2. 적용 범위

재질표기는 단일재질 기준 플라스틱 부품 100g 이상, 고무 부품 200g 이상인 부품에 적용한다.

3. 요구 사항

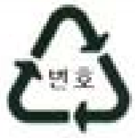
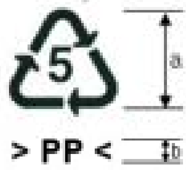
상기 적용 범위에 해당하는 부품에 한해 아래 재질표기 방법에 따라 재질표기를 실시하도록 한다.

4. 재질표기 방법

4.1 재질표기 공통 사항

1) 일반 부품과 용기류 부품의 표기법 구분

표 1. 일반 부품과 용기류 부품의 표기

<table>
<tr><th>구분</th><th>일반부품</th><th colspan="4">용기류(Reservoir, Tank)</th></tr>
<tr><td>표기법</td><td>>재질약어<</td><td colspan="4">
>재질< 표 4 참조</td></tr>
<tr><td>예시</td><td>>PP<
>PP-GF30<
>PA66-(GF25+TD5)<</td><td colspan="4"> 
> PET < > PP <</td></tr>
<tr><td rowspan="4">크기</td><td rowspan="4">>PP< 높이
문자두께
-높이: 기본 10 ㎜, *최소 3 ㎜ 이상
-문자두께: 문자 높이의 10% 이상
-양각/음각 깊이: 0.5 ㎜ 이상</td><td rowspan="4"></td><td>용량**</td><td>a(㎜)</td><td>b(㎜)</td></tr>
<tr><td>1L 미만</td><td>10</td><td>3</td></tr>
<tr><td>1~4L</td><td>20</td><td>5</td></tr>
<tr><td>4L 초과</td><td>30</td><td>10</td></tr>
</table>

* 최소 크기 : 재질 표기가 부품 품질(형상/성능)에 영향을 미치는 경우에 한하여 적용

2) 부품이 차량에 장착된 상태 또는 완전 해체(탈거 및 재질별 분리)된 상태에서 쉽게 식별이 가능한 위치에 표기한다.
3) 마모, 훼손, 오염 등에 의해 식별이 어렵지 않은 위치에 표기하여 차량의 폐차 단계에서도 식별이 가능하도록 표기한다.
4) 모든 문자는 대문자로 표기하며 문자와 문자, 문자와 기호 사이 공백이 없도록 표기한다.
5) 부품의 재질표기는 금형에 각인하는 것(양각, 음악)을 원칙으로 하되, 부득이한 경우 인쇄, 낙인성형, 레이저 표식, STAMP 등의 방법을 사용할 수 있다.
6) 부품번호, 제조업체, 제조일자 등을 표시하는 경우, 가능한 재질 표기와 함께 표기한다.

4.2 부품 타입별 재질 표기 방법(표기 예는 <부록 1> 참조)

표 2. 부품 타입별 재질 표기 방법

<table>
<tr><td colspan="4">1) 단일 재질로 분리가 가능한 복합 부품</td></tr>
<tr><td>표기법</td><td colspan="3">각 단일 부품에 재질 표기 할 것</td></tr>
<tr><td>부품예</td><td colspan="3">플라스틱 사출 부품</td></tr>
<tr><td>표기예</td><td colspan="3"><부록 1> 참조</td></tr>
<tr><td colspan="4">2) 단일 재질로 분리가 불가능한 복합 부품</td></tr>
<tr><td colspan="4">2-1) 각 단일 재질이 적층/융착/접착 되어 있는 부품</td></tr>
<tr><td>표기법</td><td colspan="3">① 각 단일 재질의 명칭에 따른 재질을 한 곳에 표기하거나
② 육안으로 확인 가능한 순서로 각 재료의 약어를 ','로 구분하여 표기함.</td></tr>
<tr><td>부품예</td><td colspan="3">인슐레이션패드, 대쉬아이소패드, 러기지 트림 등의 PAD, PNL 류 (스킨-흡음-차음/스킨-기재 등의 층 구조로 이루어진 부품), 이중사출 부품
(단, 이중사출부품, 클러스터, 램프류 등과 같이 외관상의 문제로 각 재질에 표기가 어려운 부품의 경우는 표기법①과 같이 표기할 것)</td></tr>
<tr><td>표기예</td><td>SKIN >PET<
LAYER1 >PUR<
LATER2 >GW<
또는
>PET,PUR,GW<</td><td>SKIN >PES<
CORE >PU<
또는
>PES,PU<</td><td>LENS >PC<
BEZEL >PC<
HOUSING >ABS<</td></tr>
<tr><td colspan="4">2-2) 각 단일 재질이 코팅/라미네이팅 되어 있는 부품 및 고무 부품</td></tr>
<tr><td>표기법</td><td colspan="3">육안으로 확인 가능한 순서로 각 재료의 약어를 ','로 구분하여 표기 후 중량비가 가장 높은 재질에 밑줄 표기 할 것</td></tr>
<tr><td>부품예</td><td colspan="3">고무호스, PVC 코팅 부품 등</td></tr>
</table>

표기예	① 고무호스 >ECO,THV,FKM< ② PVC 코팅 부품 >PVC,PUR,ABS<
2-3) 금속이 안에 삽입되어 있는 부품	
표기법	재료 약어 뒤에 '(금속원소기호)'를 표기
표기예	① >NBR(Fe)< ② >PVC(Cu)< ③ >PA12(Al)<
3) 길이가 긴 부품	
표기법	가능한 여러 위치에 재질을 표기
부품예	고무호스, BELT 등

4.3 재질 타입별 재질표기 방법

표 3. 재질 타입별 재질 표기 방법

1) 단일 조성 재질(Homopolymer, copolymer, Nature polymer, Rubber)		
표기법	- >Polymer< - 재질의 약어를 '> <' 안에 표기	표 5
표기예	① Polypropylene >PP< ② acrylonitrile-butadiene-styrene plastic >ABS< ③ ethylene-propylene plastic >E/P<	
2) 블렌드(Blend)		
표기법	- >Polymer1 + Polymer2< - 각 재료의 약어를 '+' 기호로 연결하여 표기(단, 순서는 재료함량순)	표 6
표기예	① PC(60%)와 PBT(40%) 블렌드 재료 >PC+PBT< ② PP(80%)와 PE(20%) 블렌드 재료 >PP+PE< ③ PA6(90%)와 HDPE(10%) 블렌드 재료 >PA6+(PE-HD)<	
3) 열가소성 탄성체(TPE, Thermoplastic Elastomer)		
표기법	- >열가소성 재료의 약어< - TPO, TPU, TPV, TPA, TPC, TPS, TPV, TPZ 등	표 9
표기예	① Olefin 계 TPE >TPO< ②Urethane 계 TPE >TPU<	
4) 첨가제(충진제 또는 보강제) 함유 재료		
표기법	- >주 재료의 약어 - 첨가제 약어 및 함량< - 주 재료의 약어와 첨가제 약어를 '-' 기호로 연결하고 첨가제의 함량을 표기 - 두 개 이상의 첨가제를 사용할 경우 첨가제는 '+' 기호로 연결	표 10, 표 11

표기예	① 단일 첨가제 - PP(주 재료)에 Glass fibre(GF) 30% 함유 재질 >PP-GF30< ② 두 개 이상의 첨가제(첨가제 함량 구분 가능) - PP(주 재료)에 Glass fibre 35%, Talc powder(TD) 15% 함유 재질 >PP-(GF35+TD15)< ③ 두 개 이상의 첨가제(첨가제 함량 구분 불가) - PP(주 재료)에 Glass fibre 와 Talc powder 의 합이 30% 함유 재질 >PP-(GF+TD)30<	
5) 가소제 (Plasticizer) *		
표기법	- >주 재료의 약어 - P(가소제 약어)< - 가소제를 의미하는 기호는 'P'로 표기하고 가소제의 코드를 "( )"안에 삽입 후 주 재료와 '-' 기호로 연결	표 12
표기예	① DBP 가소제가 포함된 PC >PC-P(DBP)<	
6) 난연제 (Flame Retardant) *		
표기법	- >주 재료의 약어 - FR(난연제 약어)< - 난연제를 의미하는 기호는 FR 로 표기하고 난연제의 약어를 "( )"안에 삽입 후 주 재료와 '-' 기호로 연결	표 13
표기예	① ISO 1043-4 의 52 호 난연제가 함유된 PA >PA-FR(52)<	

* 중국 현지 생산차에 한해 적용 (16 년 1 월 이후 신차, 18 년 1 월 이후 양산차)

5. 재질의 기호 및 약어

5.1 용기류(Reservoir, Tank) 재질 기호

표 4. 용기류 재질 기호

재질	기호	재질	기호
Polyethylene terephthalate	1 > PET <	Polypropylene	5 > PP <
High density polyethylene	2 > PE-HD <	Polystyrene	6 > PS <

Vinyl/ Polyvinyl chloride	 > PVC <	Others	 > OTHER <
Low density polyethylene	 > PE-LD <		

5.2 단일 고분자, 공중합체 및 천연 고분자 재질 약어

표 5. 단일 고분자, 공중합체 및 천연 고분자 재질 약어

약어	재료의 명칭
AB	acrylonitrile-butadiene plastic
ABAK	acrylonitrile-butadiene-acrylate plastic: preferred term for ABA
ABS	acrylonitrile-butadiene-styrene plastic
ACS	acrylonitrile-chlorinated ployethlene-stryene: preferred term for ACPES
AEPDS	acrylonitrile-(ethylene-propylene-diene)-styrene plastic: preferred term for AEPDMS
AMMA	acrylonitrile-methyl methacrylate plastic
ASA	acrylonitrile-styrene-acrylate plastic
CA	cellulose Acetate
CAB	cellulose Acetate Butyrate
CAP	cellulose Acetate Propionate
CEF	cellulose formaldehyde
CF	cresol-formaldehyde resin
CMC	carboxymethyl Cellulose
CN	cellulose Nitrate
COC	cycloolefin copolymer
CP	cellulose Propionate
CTA	cellulose Triacetate
EAA	ethylene-acrylic acid plastic
EBAK	ethylene-butyl acrylate plastic: preferred term for EBA
EC	ethyl Cellulose

EEAK	ethylene-ethyl acrylate plastic: preferred tern for EEA
EMA	ethylene-methacrylic acid plastic
EP	epoxide: epoxy resin or plastic
E/P	ethylene-propylene plastic: preferred term for EPM
ETFE	ethylene-tetrafluoroethylene plastic
EVAC [1)]	ethylene-vinyl acetate plastic: preferred term for EVA
EVOH	ethylene-vinyl alcohol plastic
FEP	perfluoro(ethylene-propylene) plastic: preferred term for PFEP
FF	furan-formaldehyde resin
HBV	poly(3-hydroxybutyrate)-co-(3-hydroxyvalerate)
LCP	liquid-crystal polymer
MABS	methyl methacrylate-acrylonitrile-butadiene-styrene plastic
MBS	methyl methacrylate-butadiene-styrene plastic
MC	methyl cellulose
MF	melamine-formaldehyde resin
MP	malamine-phenol resin
MSAN	α-methylstyrene-acrylonitrile plastic
PA	Polyamide
PA4T	homopolymer based on tetramethylenediamine and terephthalic acid
PA6	homopolymer based on ε-caprolactam
PA66	homopolymer based on hexamethylenediamine and adipic acid
PA69	homopolymer based on hexamethylenediamine and azelaic acid
PA610	homopolymer based on hexamethylenediamine and sebacic acid
PA612	homopolymer based on hexamethylenediamine and dodecanedioic acida
PA6T	homopolymer based on hexamethylenediamine and terephthalic acid
PA9T	homopolymer based on nonamethylenediamine and terephthalic acid
PA11	homopolymer based on 11-aminoundecanoic acid
PA12	homopolymer based on ω-aminododecanoic acid or on laurolactam
PAMXD6	homopolymer based on m-xylylenediamine and adipic acid
PA46	homopolymer based on tetramethylenediamine and adipic acid
PA1212	homopolymer based on dodecanediamine and dodecanedioic acid
PA66/610	copolymers based on hexamethylenediamine, adipic acid and sebacic acid
PA6/12	copolymers based on ε-caprolactam and laurolactam

PA12/IPDI	copolymers based on laurolactam, isophoronediamine and isophthalic acid
PA46/6	copolymers based on tetramethylenediamine, adipic acid and ε-caprolactam
PA4T/6T	copolymers based on tetramethylenediamine, terephthalic acid and hexamethylenediamine
PA6T/MPMDT	copolymers based on hexamethylenediamine, terephthalic acid and methylpentamethylenediamine
PA6T/66	copolymers based on hexamethylenediamine, terephthalic acid and adipic acid
PA6T/6I	copolymers based on hexamethylenediamine, terephthalic acid and isophthalic acid
PA6T/6I/66	copolymers based on hexamethylenediamine, terephthalic acid, isophthalic acid and adipic acid
PA66/6I	copolymers based on hexamethylenediamine, adipic acid and isophthalic acid
PANDT/INDT	copolymers based on 1,6-diamino-2,2,4-trimethylenehexane, 1,6-diamino-2,4,4-trimethylenehexane and terephthalic acid
PA12/IPDI	copolymers based on laurolactam, isophoronediamine and isophthalic acid
PAA	poly(acrylic acid)
PAEK	Polyacryletherketone
PAI	polyamidimide
PAK	polyacrylate
PAN	polyacrylonitrile
PAR	polyarylate
PARA	poly(aryl amide)
PB	polybutene
PBAK	polybuthyl acrylate
PBD	1,2-polybutadiene
PBN	poly(butylene naphthalate)
PBS	poly(butylene succinate)
PBSA	poly(butylene succinate adipate)
PBT	poly(butylene terephthlate)
PC	polycarbonate

PCCE	poly(cyclohexylene dimethylene cyclohexanedicarboxylate)
PCO	polycycloolefin
PCL	polycaprolactone
PCT	poly(cyclohexylene dimethylene terephthalate)
PCTFE	polychlorotrifluoroethylene
PDAP	poly(diallyl phthalate)
PDCPD	polydicyclopentadiene
PE	polyethylene
PE-C [2)]	polyethylene, chlorinated; preferred term for CPE
PE-HD	polyethylene, high density; preferred term for HDPE
PE-LD	polyethylene, low density; preferred term for LDPE
PE-LLD	polyethylene, linear low density; preferred term for LLDPE
PE-MD	polyethylene, medium density; preferred term for MDPE
PE-UHMW	polyethylene, ultra high molecular weight; preferred term for UHMWPE
PE-VLD	polyethylene, very low density; preferred term for VLDPE
PEC	polyestercarbonate
PEEKK	Polyetheretherketoneketone
PEEST	polyetherester
PEI	polyetherimide
PEK	polyetherketone
PEN	poly(ethylene naphthalate)
PEOX	poly(ethylene oxide)
PES	poly(ethylene succinate)
PESTUR	polyesterurethane
PESU	polyethersulfone
PET	poly(ethylene terephthalate)
PEUR	polyetherurethane
PF	phenol-formaldehyde resin
PFA	perfluoro alkoxyl alkane resin
PHA	polyhydroxyalkanoate
PHB	poly(3-hydroxybutyrate)
PI	polyimide
PIB [3)]	polyisobuthylene

PIR	polyisocyanurate
PK	polyketone
PLA	poly(lactic acid)
PMI	polymethacrylimide
PMMA	poly(methyl methacrylate)
PMMI	poly-*N*-methylmethacrylimide
PMP	poly-4-methylpent-1-ene
PMS	poly-α-methylstyrene
POM	polyoxymethylene; polyacetal; polyformaldehyde
PP	polypropylene
PP-E	polypropylene, expandable; preferred term for EPP
PP-HI	polypropylene, high impact; preferred term for HIPP
PPE	poly(phenylene ether)
PPOX	poly(propylene oxide)
MPPO	Modified poly(propylene oxide)
PPS	poly(phenylene sulfide)
PPSU	poly(phenylene sulfone)
PS	polystyrene
PS-E	polystyrene, expandable; preferred term for EPS
PS-HI	polystyrene, high impact; preferred term for HIPS
PS-S	polystyrene, sulfonated
PSU	polysulfone
PTFE	polytetrafluoroethylene
PTT	poly(trimethylene terephthalate)
PUR	polyurethane
PUR-E	polyurethane, expandable
PVAC	poly(vinyl acetate)
PVAL	poly(vinyl alcohol); preferred term for PVOH
PVB	poly(vinyl butyral)
PVC	poly(vinyl chloride)
PVC-C	poly(vinyl chloride), chlorinated; preferred term for CPVC
PVC-U	poly(vinyl chloride), unplasticized; preferred term for UPVC
PVDC	poly(vinylidene chloride)

PVDF	poly(vinylidene fluoride)
PVF	poly(vinyl fluoride)
PVFM	poly(vinyl formal)
PVK	poly-*N*-vinylcarbazole
PVP	poly-*N*-vinylpyrrolidone
SAN	styrene-acrylonitrile plastic
SB	styrene-butadiene plastic
SI [4]	silicone plastic
SMAH	styrene-maleic anhydride plastic; preferred term for S/MA or SMA
SMS	styrene-α-methylstyrene plastic
UF	urea-formaldehyde resin
UP	unsaturated polyester resin
VCE	vinyl chloride-ethylene plastic
VCEMAK	vinyl chloride-ethylene-methyl acrylate plastic; preferred term for VCEMA
VCEVAC	vinyl chloride-ethylene-vinyl acetate plastic
VCMAK	vinyl chloride-methyl acrylate plastic; preferred term for VCMA
VCMMA	vinyl chloride-methyl methacrylate plastic
VCOAK	vinyl chloride-octyl acrylate plastic; preferred term for VCOA
VCVAC	vinyl chloride-vinyl acetate plastic
VCVDC	vinyl chloride-vinylidene chloride plastic
VE	vinyl ester resin

1), 2), 3), 4) 고무재료의 경우 표 8의 표기법을 따를 것

5.3 블렌드(Blend) 재질 약어

표 6. 블렌드 재질 약어

기호	명칭/설명
PA66+PPE	PA66 and poly(phenylene ether)
PC+PBT	polycarbonate and poly(butylene terephthlate)
PBT+PET	poly(butylene terephthlate) and poly(ethylene terephthalate)
PC+ABS	polycarbonate and acrylonitrile/butadiene/styrene
ABS+PMMA	acrylonitrile-butadiene-styrene plastic and poly(methyl methacrylate)
POM+PUR	polyoxymethylene and polyurethane

PPE+PS	poly(phenylene ether) and polystyrene
PVC+PUR	poly(vinyl chloride) and polyurethane

5.4 인조섬유 재질 약어

표 7. 인조섬유 재질 약어

명칭	기호	명칭	기호
acetate	CA	lyocell	CLY
acrylic	PAN	Metal fiber (금속섬유)	MTF
alginate	ALG	modacrylic	MAC
aramid	AR	modal	CMD
Carbon (탄소섬유)	CF	polyamide (or nylon)	PA
chlorofibre	CLF	polyester	PES
cupro	CUP	polyethylene	PE
elastane (or spandex)	EL	polyimide	PI
elastodienea	ED	polylactide (or PLA)	PLA
elastolefin (or lastol)	EOL	polypropylene	PP
elastomultiester (or elasterell-p)	ELE	triacetate	CTA
fluorofibre	PTFE	vinylal	PVAL
Glass(유리섬유)	GF	viscose (or rayon)	CV

5.5 고무재료 재질 약어

표 8. 고무재료 재질 약어

그룹	기호	명칭/설명
M	ACM	Copolymer of ethyl acrylate (or other acrylates) with low monomeric content for easier vulcanization
	AEM	Copolymer of ethyl acrylate (or other acrylates) and ethylene
	ANM	Copolymer of ethyl acrylate (or other acrylates) and acrylonitrile
	CM	Chloropolyethylene
	CSM	Chlorosulfonylpolyethylene
	EBM	Ethylene-butene copolymer
	EOM	Ethylene-octene copolymer
	EPDM	Ethylene/Propylene/Diene(polymer)
	EPM	Ethylene-Propylene copolymer
	EVM	Ethylene-vinyl acetate copolymer
	FEPM	Copolymer of tetrafluoroethylene and propylene
	FFKM	Perfluoro rubber in which all substituent groups on the polymer chain are fluoro, perfluoroalkyl or perfluoroalkoxy groups
	FKM	Fluoro rubber having substituent fluoro, perfluoroalkyl or perfluoroalkoxy groups on the polymer chain
	IM	Polyisobutene
	NBM	Fully hydrogenerated acrylonitrile-butadiene copolymer
	SEBM	Styrene-ethylene-butene terpolymer
	SEPM	Styrene-ethylene-propylene terpolymer
O	CO	Polychloromethyloxirane
	ECO	Copolymer of ethylene oxide(oxirane) and chloromethyloxirane
	GCO	Copolymer of epichlorohydrin and allyl glycidyl ether
	GECO	Terpolymer of epichlorohydrin-ethylene oxide-allyl glycidyl ether
	GPO	Copolymer of propylene oxide and allyl glycidyl ether
	MVO	Silicon ruuber with methyl- and vinyl- group
	TEO	Thermoplastic elastomer "polyolefinic"
Q	FMQ	Silicon rubber having both methyl and fluorine substituent groups on the polymer chain
	FVMQ	Silicon rubber having methyl, vinyl and fluorine substituent groups on the polymer chain

	MQ	Silicon rubber having only methyl substituent groups on the polymer chain, such as dimethyl polysiloxane
	PMQ	Silicon rubber having both methyl and phenyl substituent groups on the polymer chain
	PVMQ	Silicon rubber having both methyl and vinyl substituent groups on the polymer chain
	VMQ	Silicon rubber having methyl and vinyl substituent groups on the polymer chain
R	ABR	Acrylate-butadiene rubber
	BR	Butadiene rubber
	CR	Chloroprene rubber
	ENR	Epoxidized natural rubber
	HNBR	Hydrogenated NBR
	IIR	Isobutene-isoprene rubber
	IR	Isoprene rubber, synthetic
	MSBR	α-methylstyrene-butadiene rubber
	NBIR	Acrylonitrile-butadiene-isoprene rubber
	NBR	Acrylonitrile-butadiene rubber
	NIR	Acrylonitrile-isoprene rubber
	NOR	Norbornene rubber
	NR	Natural rubber
	PBR	Vinylpyridine-butadiene rubber
	PSBR	Vinylpyridine-styrene-butadiene rubber
	SBR	Styrene-butadiene rubber
	SIBR	Styrene-isoprene-butadiene rubber
R	XBR	Carboxylic-butadiene rubber
	XCR	Carboxylic-chloroprene rubber
	XNBR	Carboxylic-acrylonitrile-butadiene rubber
	XSBR	Carboxylic-styrene-butadiene rubber
	BIIR	Bromo-isobutene-isoprene rubber
	CIIR	Chloro-isobutene-isoprene rubber
T	OT	A rubber having either a $-CH_2-CH_2-O-CH_2-O-CH_2-CH_2-$ group or occasionally an R group, where R is an alphatic hydrocarbon, not usually $-CH_2-CH_2-$, between the polysulfide linkages in the polymer chain.

	EOT	A rubber having a $-CH_2-CH_2-O-CH_2-O-CH_2-CH_2-$ group and R group which are usually $-CH_2-CH_2-$ but occasionally other alphatic groups between the polysulfide linkages in the polymer chain.
U	AFMU	Terpolymer of tetrafluoroethylene, trifluoronitrosomethane and nitrosoperfluorobutyric acid
	AU	Polyester urethane
	EU	Polyester urethane
Z	FZ	A rubber having a -P=N- chain and having fluoroalkoxy groups attached to the phosphorus atoms in the chain
	PZ	A rubber having a -P=N- chain and having aryloxy(phenoxy and substituted phenoxy) groups attached to the phosphorus atoms in the chain

5.6 열가소성 탄성체(TPE, Thermoplastic Elastomer) 재질 약어

표 9. 열가소성 탄성체 재질 약어

그룹	기호	명칭/설명
TPA	Polyamide thermoplastic elastomer, comprising a block copolymer of alternating hard and soft segments with amide chemical linkages in the hard blocks and ether and/or ester linkages in the soft blocks.	
	TPA-EE	Soft segment with both ether and ester linkages
	TPA-ES	Polyester soft segment
	TPA-ET	Polyether soft segment
TPC	Copolyester thermoplastic elastomer, consisting a block copolymer of alternating hard and soft segments, chemical linkages in the main chain being ether and/or ester.	
	TPC-EE	Soft segment with ester and ether linkages
	TPC-ES	Polyester soft segment
	TPC-ET	Polyether soft segment
TPO	Olefinic thermoplastic elastomer, consisting of blend of a polyolefin and a conventional rubber, the rubber phase in the blend having little or no crosslinking. The thermoplastic and the rubber type shall be listed in decreasing order of abundance in the TPO.	
	TPO-(EPDM+PP)	Blend of ethylene-propylene-diene terpolymer with polypropylene, with no or little crosslinking of the EPDM phase, the amount of EPDM present being greater than that of PP

TPS		Styrenic thermoplastic elastomer, consisting of at least a triblock copolymer of styrene and a specific diene, where the two end blocks (hard blocks) are polystyrene and the internal block (soft block or blocks) is a polydiene or hydrogenated polydiene.
	TPS-SBS	Block copolymer of styrene and butadiene
	TPS-SEBS	Polystyrene-poly(ethylene-butadiene-)-polystyrene
	TPS-SEPS	Polystyrene-poly(ethylene-propylene-)-polystyrene
	TPS-SIS	Block copolymer of styrene and isoprene
TPU		Urethane thermoplastic elastomer, consisting of a block copolymer of alternating hard and soft segments with urethane chemical linkages in the hard blocks and ether, ester or carbonate linkages or mixtures of them in the soft blocks.
	TPU-ARES	Aromatic hard segment, polyester soft segment
	TPU-ARET	Aromatic hard segment, polyether soft segment
	TPU-AREE	Aromatic hard segment, soft segment with ester and ether linkages
	TPU-ARCE	Aromatic hard segment, polycarbonate soft segment
	TPU-ARCL	Aromatic hard segment, polycaprolactone soft segment
	TPU-ALES	Aliphatic hard segment, polyester soft segment
	TPU-ALET	Aliphatic hard segment, polyester soft segment
TPV		Thermoplastic rubber vulcanizate consisting of a blend of a thermoplastic material and a conventional rubber in which the rubber has been crosslinked by the process of dynamic vulcanization during the blending and mixing step. The abbreviation for the rubber shall precede that for the thermoplastic
	TPV-(EPDM+PP)	Combination of EPDM and polypropylene in which the EPDM phase is highly crosslinked and finely dispersed in a continuous polypropylene phase
	TPV-(NBR+PP)	Combination of acrylonitrile-butadiene rubber and polypropylene in which the NBR phase is highly crosslinked and finely dispersed in a continuous polypropylene phase
	TPV-(NR+PP)	Combination of natural rubber and polypropylene in which the NR phase is highly crosslinked and finely dispersed in a continuous polypropylene phase
	TPV-(ENR+PP)	Combination of epoxidized natural rubber and polypropylene in which the ENR phase is highly crosslinked and finely dispersed in a continuous polypropylene phase

	TPV-(IIR+PP)	Combination of butyl rubber and polypropylene in which the IIR phase is highly crosslinked and finely dispersed in a continuous polypropylene phase
TPZ	Unclassified thermoplastic elastomer comprising any composition or structure other than those grouped in TPA, TPC, TPO, TPS, TPU and TPV.	
	TPZ-(NBR+PVC)	Blend of acrylonitrile-butadiene rubber and poly(vinyl chloride)

5.7 첨가제(충전제 및 보강제) 재질 약어

표 10. 첨가제 약어 표기법

표기법	'물질기호' 와 '형태기호'를 결합하여 표기	
표기예	① 유리(G) 섬유(F) GF ② 유리(G) 비드(B) GB ③ 미네랄(M) 휘스커(H) MH	④ 미네랄(M) 파우더(D) MD ⑤ 탈크(T) 파우더(D) TD ⑥ 카본(C) 나노튜브(NT) CNT

표 11. 첨가제 약어 기호

기호	재료	기호	형태/구조
A	aramid	B	beads, spheres, balls, bubble
B	boron	C	chips, cuttings
C	carbon	CM	chopped-strand mat
D	alumina trihydrate	D	fines, powder
E	clay	EM	continuous (endless) strand mat
G	glass	F	fibre
K	calcium carbonate	G	ground
L	cellulose	H	whisker
M	mineral	K	knitted fabric
ME	metal	L	layer
N	natural organic (cotton, sisal, hemp, flax, etc.)	LF	long fabric
P	mica	M	mat (thick)
Q	silica	N	non-woven(fabric, thin)
S	synthetic organic (e.g. finely devided PTFE, polyimides or thermoset resins)	NF	Nanofibres
T	talc	NT	nanotubes
W	wood	P	paper

X	not specified	R	roving
Z	others	S	flake
		T	twisted or braided fabric, cord, tube
		V	veneer
		W	woven fabric
		X	not specified
		Y	yarn
		Z	others

5.8 가소제 (Plasticizers) 약어

표 12. 가소제 약어 기호

약어	관용명	IUPAC 명	CAS-RN
ASE	alkylsulfonic acid ester	alkanesuifonates or aikyi alkanesulfonates	not known
BAR	butyl o-acetylricinoleate	butyl(R)-12-acetoxyoleate	140-04-5
BBP	benzyl butyl phthalate	동일	85-68-7
BCHP	butyl cyclohexyl phthalate	동일	54-64-0
BNP	butyl nonyl phthalate	동일	not known
BOA	benzyl octyl adipate	Benzyl 2-ethylhexyl adipate	3089-55-2
BOP	butyl octyl phthalate	butyl 2-ethylhexyl phthalate	85-69-8
BST	butyl stearate	동일	123-95-5
DBA	dibutyl adipate	동일	105-99-7
DBEP	di-(2-butoxyethyl) phthalate	bis(2-butoxyethyl) phthalate	117-83-9
DBF	dibutyl fumarate	동일	105-75-9
DBM	dibutyl maleate	동일	105-76-0
DBP	dibutyl phthalate	동일	84-74-2
DBS	dibutyl sebacate	동일	109-43-3
DBZ	dibutyl azelate	동일	2917-73-9
DCHP	dicyclohexyl phthalate	동일	84-61-7
DCP	dicapryl phthalate	bis(1-methylheptyl) phthalate	131-15-7
DDP	didecyl phthalate	동일	84-77-5
DEGDB	Diethylene glycol dibenzoate	oxydiethylene dibenzoate	120-55-8
DEP	diethyl phthalate	동일	84-66-2
DHP	diheptyl phthalate	동일	3468-21-3

DHXP	dihexyl phthalate	동일	84-75-3
DIBA	diisobutyl adipate	동일	141-04-8
DIBM	diisobutyl maleate	동일	14234-82-3
DIBP	diisobutyl phthalate	동일	84-69-5
DIDA	diisodecyl adipate	-	27178-16-1
DIDP	diisodecyl phthalate	-	26761-40-0
DIHP	diisoheptyl phthalate	-	41451-28-9
DIHXP	diisohexyl phthalate	diisohexyl phthalate	71850-09-4
DINA	diisononyl adipate	-	33703-08-1
DINP	diisononyl phthalate	-	28553-12-0
DIOA	diisooctyl adipate	-	1330-86-5
DIOM	diisooctyl maleate	-	1330-76-3
DIOP	diisooctyl phthalate	-	27554-26-3
DIOS	diisooctyl sebacate	-	27214-90-0
DIOZ	diisooctyl azelate	-	26544-17-2
DIPP	diisopentyl phthalate	동일	605-50-5
DMEP	di-(2methyloxyethyl) phthalate	bis(2-methoxyethyl) phthalate	117-82-8
DMP	dimethyl phthalate	동일	131-11-3
DMS	dimethyl sebacate	동일	106-79-6
DNF	dinonyl fumarate	동일	2787-64-6
DNM	dinonyl maleate	동일	2787-64-6
DNOP	di-n-octyl phthalate	dioctyl phthalate	117-84-0
DNP	dinonyl phthalate	dinonyl phthalate	14103-61-8
DNS	dinonyl sebacate	dinonyl sebacate	4121-16-8
DOA	dioctyl 3) adipate	bis(2-ethylhexyl)3) adipate	103-23-1
DOIP	dioctyl isophthalate	bis(2-ethylhexyl) isophthalate	137-89-3
DOP	dioctyl phthalate	bis(2-ethylhexyl) phthalate	117-81-7
DOS	dioctyl sebacate	bis(2-ethylhexyl) sebacate	122-62-3
DOTP	dioctyl terephthalate	bis(2-ethylhexyl) terephthalate	6422-86-2
DOZ	dioctyl azelate	bis(Z-ethylhexyl) azelate	2064-80-4
DPCF	diphenyl cresyl Phosphate	diphenyl x-tolyl orthophosphate, where x denotes o, m, p or	26444-49-5

		mixture	
DPGDB	di-x-propylene glycol dibenzoate	-	not known
DPOF	diphenyl octyl Phosphate	2-ethylhexyl diphenyl orthophosphate or octyl diphenyl orthophosphate	1241-94-7
DPP	diphenyl phthalate	diphenyl phthalate	84-62-8
DTDP	diisotridecyl phthalate(SEE2.8)	-	27253-26-5
DUP	diundecyl phthalate	diundecyl phthalate	3648-20-2
ELO	epoxidized linseed oil	not possible	8016-11-3
ESO	epoxidized soya bean oil	not possible	8013-07-8
GTA	glycerol triacetate	glycerol triacetate	102-76-1
HNUA	heptyl nonyl undecyl adipate (= 711A)	-	not known
HNUP	heptyl nonyl undecyl phthalate (= 711P)	-	68515-42-4
HXODA	hexyl octyl decyl adipate (= 61 OA)	-	not known
HXODP	hexyl octyl decyl phthalate (= 610P)	-	68515-51-5
NUA	nonyl undecyl adipate (= 911A)	-	not known
NUP	nonyl undecyl phthalate (= 911P)	-	not known
ODA	octyl decyl adipate	decyl octyl adipate	110-29-2
ODP	octyl decyl phthalate	decyl octyl phthalate	68515-52-6
ODTM	n-octyl decyl trimellitate	decyl octyl hydrogen benzene-1,2,4-tricarboxylate	not known
PO	paraffin oil	-	8012-95-1
PPA	poly(propylene adipate)	poly(propylene adipate)	not known
PPS	poly(propylene sebacate)	-	not known
SOA	sucrose octa-acetate	sucrose octaacetate	126-14-7

TBAC	tributyl o-acetylcitrate	tributyl o-acetylcitrate	77-90-7
TBEP	tri-(2-butoxyethyl) Phosphate	tris(2-butoxyethyl) orthophosphate	78-51-3
TBP	tributyl phosphate	tributyl orthophosphate	126-73-8
TCEF	trichloroethyl phosphate	tris(2-chloroethyl) orthophosphate	115-96-8
TCF	tricresyl phosphate	tri-x-tolyl orthophosphate, where x denotes o, m, p or mixture	1330-78-5
TDBPP	tri-(2,3-dibromopropyl) phosphate	tris(2,3-dibromopropyl) orthophosphate	126-72-7
TDCPP	tri-(2,3-dichloropropyl) phosphate	tris(2,3-dichloropropyl) orthophosphate	78-43-3
TEAC	triethyl o-acetylcitrate	triethyl o-acetylcitrate	77-89-4
THFO	tetrahydrofurfuryl oleate	tetrahydrofurfuryl oleate	5420-17-7
THTM	triheptyl trimellitate	triheptyl benzene-1,2,4-tricarboxylate	1528-48-9
TIOTM	triisooctyl trimellitate	tris(6-methylheptyl) benzene-1,2,4-tricarboxylate	27251-75-8
TOF	trioctyl phosphate	tris(2-ethylhexyl) orthophosphate	78-42-2
TOPM	tetraoctyl pyromellitate	tetrakis(2-ethylhexyl) benzene-1,2,4,5-tetracarboxylate	3126-80-5
TOTM	trioctyl trimellitate	tris(2-ethylhexyl) benzene-1,2,4-tricarboxylate	89-04-3
TPP	triphenyl phosphate	triphenyl orthophosphate	115-86-6
TXF	trixylyl phosphate	tri-x,y-xylyl orthophosphate where x and y denote o, m, p or mixture	25155-23-1

5.9 난연제 (Flame Retardant) 약어

표 13. 난연제 약어 기호

그룹	코드	명칭/설명
할로겐 화합물	10	Aliphatic/alicyclic chlorinated compounds
	11	Aliphatic/alicyclic chlorinated compounds in combination with antimony compounds
	12	Aromatic chlorinated compounds
	13	Aromatic chlorinated compounds in combination with antimony compounds
	14	Aliphatic/alicyclic brominated compounds (excluding hexabromocyclododecane)
	15	Aliphatic/alicyclic brominated compounds (excluding hexabromocyclododecane) in combination with antimony compounds
	16	Aromatic brominated compounds(excluding brominated diphenyl ether and biphenyls)
	17	Aromatic brominated compounds(excluding brominated diphenyl ether and biphenyls) in combination with antimony compounds
	18	Polybrominated diphenyl ether
	19	Polybrominated diphenyl ether in combination with antimony compounds
	20	Polybrominated biphenyls
	21	Polybrominated biphenyls in combination with antimony compounds
	22	Aliphatic/alicyclic chlorinated and brominated compounds
	23	hexabromocyclododecane
	24	hexabromocyclododecane with antimony compounds
	25	Aliphatic fluorinated compounds
	26~29	Not allocated
니트로겐 화합물	30	Nitrogen compounds(confined to melamine, melamine cyanurate, urea)
	31~39	Not allocated
유기 인 화합물	40	Halogen-free organic phosphorus compounds
	41	Chlorinated organic phosphorus compounds

	42	Brominated organic phosphorus compounds
	43~49	Not allocated
무기 인 화합물	50	Ammonium orthophosphates
	51	Ammonium polyphosphates
	52	Red phosphates
	53~59	Not allocated
금속산화물 금속수산화물 금속염	60	Aluminium hydroxide
	61	Magnesium hydroxide
	62	Antimony(III) oxide
	63	Alkali-metal antimonate
	64	Magnesium/calcium carbonate hydrate
	65~69	Not allocated
보론 & 아연 화합물	70	Inorganic boron compounds
	71	Organic boron compounds
	72	Zinc borate
	73	Organic zinc compounds
	74	Not allocated
실리카 화합물	75	Inorganic silicon compounds
	76	Organic silicon compounds (silicones)
	77~79	Not allocated
	80	Graphite
	81~89	Not allocated
	90~99	Not allocated

6. 관련 규격

ISO11469, ISO 1874-1, ISO1043-1, ISO1043-2, ISO1043-4, ISO1629, ISO18064, QC/T797-2008, GB/T1844.3-2008

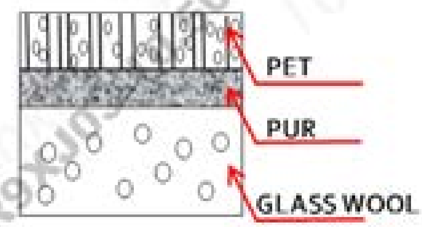
<부록 1> 부품 타입별 재질 표기 예

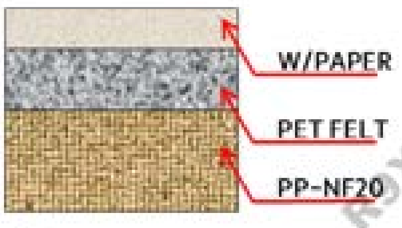
1) 단일 재질로 분리가 가능한 복합 부품
  - 표기법 : 각 단일 부품에 재질 표기 할 것
  - 표기예

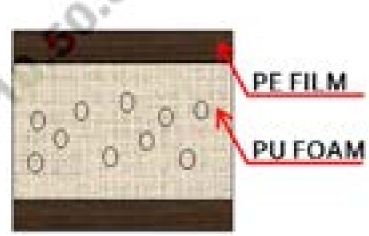
도어트림 [복합 부품]	=	>PP-GF30< 어퍼 판넬 [단일 부품]	+	>PA66-TD20< 로워 판넬 [단일 부품]	+	>ABS< 센터 판넬 [단일 부품]	+	>PP-(TD+GF)35< 가니쉬 [단일 부품]

2) 단일 재질로 분리가 불가능한 복합 부품
  - 표기법 1 : 각 단일 재질 층에 따른 재질을 표기(가장 단순한 형태의 재질약어를 사용할 것)
  - 표기법 2 : 육안으로 확인 가능한 순서로 각 재료의 약어를 ','로 구분하고 표기 후 중량비가 가장 높은 재질에 밑줄 표기

  2-1) 적층 부품 (예, PAD, PNL)
   - 표기법 1, 2 모두 표기 가능(단, 표기법 2 의 경우 밑줄은 긋지 않음)
   - 표기예







SKIN >PET< LAYER1 >PUR< LAYER2 >GW< 또는 >PET,PUR,GW<	SKIN >PES< LAYER1 >PET< LAYER2 >PP-NF20< 또는 >PES,PET,PP-NF20<	SKIN >PE< CORE >PU< 또는 >PE,PU<

※ PES, PET 같은 의미로 표기 허용함.

* CLUSTER/LAMP ASSY
  - 표기법 : 각 단일 재질의 명칭과 재질을 한 곳에 표기
  - 표기예

①

BODY >PC<
MOLD'G >ABS+PC<
BEZEL >PC<
LENS >PC<
B/COVER >PC<

②

OUT LENS >PMMA<
INR LENS >PC<
BEZEL >PC<
HOUSING >ABS<

※ 외관이 투명한 부품의 경우 한곳에 재질표기 함.

2-2) 고무 및 코팅/라미네이팅 부품
  - 표기법 2 로 표기함
  - 표기예

① 고무호스 부품

>ECO,THV,FKM<

② PVC가 코팅된 부품

중심에 ABS가 있고 그것을 PUR이 감싸고 있으며
PVC가 표면 코팅된 부품 (중량비 : ABS>PUR>PVC)

>PVC,PUR,ABS<

<부록 2> 재질별 오류 표기 및 바른표기 예

1) 단일 조성 재질(Basic Polymer)	
오류표기	바른표기
>ERP<	>E/P<
>HD-PE< >PE-HD-010<	>PE-HD<
>EVA<	>EVAC<
2) 블렌드	
>PP-PET35<	>PP+PET<
>ABS/PC< >ABS,PC< >PC ABS<	>PC+ABS<
3) 열가소성 탄성체(TPE)	
>TPE<	>TPO< >TPO-(PP+EPDM)<
>TPE+TD10<	>TPO-TD10<
4) 첨가제 함유 재질	
>PP+GF30<	>PP-GF30<
>PP,GF20< >PP+GF20%<	>PP-GF20<
>PA6-MD30%<	>PA6-MD30<
>PA66-GF15%<	>PA66-GF15<
>PP-TD<	>PP-TD10*< (*실제 함량에 맞게 숫자 표기)
>PP-TALC20%< >PP+TD20< >PP+T20<	>PP-TD20<
>PA66-(GF17+M21)<	>PA66-(GF17+MD21)<
>PP+EPDM-T35<	>PP+EPDM-TD35<
>PP+EPR-TD10<	>PP+E/P-TD10<
>PP+EPR-(MH+TD)14<	>PP+E/P-(MH+TD)14<
>PP+GF25+MF15< >PP-GF25-MF15<	>PP-(GF25+MD15)<

>PPF<	>PP-TD10*< >PP+E/P-TD10*< (*실제 함량에 맞게 숫자 표기)
>PA6+ABS+ME/GF40<	>PA6+ABS-(GF20+MD20)<
>PP+GB< >PP+G/B< >PP-G-B<	>PP-GB20*< (*실제 함량에 맞게 숫자 표기)
>PP-T(GB+TD)10< >PP(G/B+TD)<	>PP-(GB+TD)10<
>>PP-T40<<	>PP-TD40<
>PBT GF 30%<	>PBT-GF30<